

National College of Ireland

Class Assessment Cover Sheet

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Programme: M.Sc. In Data Analytics **Year:** 2026-2027

Module: Machine Learning and Data Mining

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Submission Due Date: 27 July 2026

Project Title: Explainable Financial Health Scoring with TabTransformer, SHAP, and LLM-Generated Analyst Narratives

Word Count: 8190

Academic integrity certificate: 6IUoUeD6C8

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Date: 27 july 2026

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AI Acknowledgement Supplement

Deep Learning and Generative AI

Explainable Financial Health Scoring with TabTransformer, SHAP, and LLM-Generated Analyst

Narratives

Your Name/Student Number	Course	Date
Subhash Lora Jat/25171496 Aditya Tomar / 25124757 Smit Gandubhai Savaliya / 25122371	M.Sc. in Data Analytics	27 July 2026

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
1. ChatGPT	To understand the project related concept	ChatGPT

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. One table should be used for each tool used.

ChatGPT
To understand the project related concept

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence

What are the theoretical differences between SHAP TreeExplainer and DeepSHAP? When should each be used and what are the computational trade-offs for production financial models?

TreeExplainer and **DeepSHAP (DeepExplainer)** both produce SHAP values based on **Shapley values from cooperative game theory**, but they are designed for **different model families** and use different algorithms.

Aspect	TreeExplainer	DeepSHAP (DeepExplainer)
Target models	Decision trees, Random Forest, XGBoost, LightGBM, CatBoost	Deep neural networks (TensorFlow, PyTorch)
Theory	Exact Tree SHAP algorithm	Approximate SHAP using DeepLIFT rules
Explanation accuracy	Exact (for supported tree models)	Approximate
Computational cost	Low	Higher
Production suitability	Excellent	Moderate
Background dataset	Optional in some modes	Required

...

+ Ask anything



Think



Explainable Financial Health Scoring with TabTransformer, SHAP, and LLM-Generated Analyst Narratives

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Abstract—Bankers, auditors, and portfolio managers regularly encounter the need to classify whether a firm is “safe,” “risky,” or “distressed.” Two challenges have prevented machine learning from systematically supporting such decisions to date: (a) the highest accuracy machines are black boxes, and (b) they produce un-narrative numeric vectors. We describe an end-to-end pipeline to mitigate both. For that, we create an annotated 9,284 American public-company annual observations sourced from SimFin. We assign Altman’s Z’-Score as an proxy of sorts of our ‘ground truth’ for three financial-health categories. We use Altman’s Z’-Score as an proxy of sorts of our ‘ground truth’ for three financial states. Thereafter train an Logistic Regression, XGBoost, and Multi-Layer Perceptron against a TabTransformer all with set fixed random seed and stratified training by company to avoid silently unfair evaluation, keeping our training and test data the same — XGBoost yields best F1 (macro 0.847), winning out the deeper tabular models too. After that, we explain the XGBoost predictions using TreeSHAP, and we convert the SHAP contributions into analyst-style narratives via GPT-4o-mini, but only under a structured prompt with six rules. For 100 narratives from the test set, our automatic checks show 98% feature faithfulness, and also 0% numeric hallucination. Finally we also do a human study with three raters, on a stratified sample of 30 narratives, and the scores for Clarity, Accuracy, and Usefulness are all above 4.4 out of 5 on a Likert scale. Overall, the pipeline suggests that the currently-fashionable trade-off between accuracy and interpretability is kinda a false dilemma: you can get state-of-the-art classification plus audit-ready natural-language explanations at the same time, basically with no meaningful extra expense (USD 0.02 for the full narrative run).

Index Terms—financial distress prediction, XGBoost, TabTransformer, SHAP, explainable AI, large language models, GPT-4o-mini, narrative generation.

I. INTRODUCTION

Each quarter one of the major challenges faced by every commercial bank, rating agency, and auditing firm is to identify healthy companies from troubled companies; at risk. The process of analysing all those companies (at scale) by hand is both painful and expensive. Methods of ML for it are fast and cheap, and those models that do best for tabular financial data (gradient boosted trees and deep tabular transformers) are black boxes to the auditors, audit committee members, and

ultimate investment committee that would have to eventually approve. Outcome is a standoff: models that work best don’t get trust and most trusted methods, judgmental work and credit ratios are doing less than they could.

Therefore, A new invention was needed to be able to output per instance explanation as trees already know how to explain the predictions [1] (now with a unit price which can be as cheap as \$0.01 for LLMs equal volume & cost), LLMs have the ability to add a cheap explanation layer on top of these explanations. If a credit officer get a summary about a prediction from SHAP about a company, as little as 15 seconds to readable. Each one can not solve the black-box problem but combined gives a reproducible pipeline solution of explainer, both human verifiable with explanation technique itself (SHAP), easily readable and transparency at its core.

This paper sets up and validates this combined pipeline. It asks two research questions:

RQ1 — Modelling. On the same features and same company-level splits, are deep tabular models (Multi-Layer Perceptron, TabTransformer) superior to gradient-boosted trees (XGBoost) for the financial-health classification task on a multi-class classification problem?

RQ2 — Narratives. Can a general-purpose LLM be used via a structured prompt to generate from SHAP values analytic narratives faithfully to the model, that do not contain numeric hallucination and are helpful for human users?

We answer both empirically. For RQ1 we report an unambiguously negative finding: our dataset ($n = 9,284$ company-years, $n = 23$ engineered features) results in XGBoost outperforming the two neural architectures by 5.8 and 9.3 macro-F1 points, respectively, at one sixteenth and one fortieth of the training computation for a single feature. For RQ2 we report a positive result: under a six-rule prompt using GPT-4o-mini at 0.4 temperature, 98 percent of the 100 outputted narratives accurately named all three top-SHAP features without a single hallucinations of any number, and three independent human reviewers rate output as 4.49–4.50 on Clarity, Accuracy and Usefulness.

The paper’s impact is three-fold: First in a comparative, hon-

est and transparent review of deep tabular as against classical model-level methods in our dataset. Second, we demonstrate a practical approach for eliminating numeric hallucinations and for generating helpful narrative insights from any high-stakes decision model for any task and problem and finally, we create an inspectable and replicable three-level pipeline (Prediction, Explanation, Narratives) which effectively separates The Model auditability concerns from any User-Interface-level interaction or user-oriented insight delivery; We also include complete python code as well as the training data for The Model including all subsequent cleaning, feature-transformation processes, TrainedModel.

The paper structure proceeds as follows. Section II describes related work in these three areas; Section III presents our use of the CRIP-DM methodology; Section IV quantifies our models' performance and reader evaluations of the generated narratives; Section V discusses implications in regulated contexts, and Section VI concludes and suggests future research.

II. RELATED WORK

Our pipeline sits on the intersection of these three research fields: corporate distress prediction, deep learning for tabular data and explanation generation using LLMs. We first examine each research field and then analyze the advantages and disadvantages that influenced our design choices.

A. Financial-Distress Prediction

Original Altman's 1968 Z-Score [2] employed five financial ratios within a linear discriminant function with the goal of predicting if a company would fail within two years. The primary utility of the model is its capability of summarizing a company's financial statements into a single, intuitively interpretable number, partitioned into ranges of values. Logistic regression by Ohlson [3] and probit by Zmijewski [4] superseded the traditional discriminant analysis approach with probabilistic models that, however, maintained the ratio-based feature set.

Beaver's earlier univariate study [5] highlights the two most significant predicting ratios (in terms of explanatory power), namely cash-flow-to-debt and net-income-to-total-assets, both of which are part of the newer multivariate frameworks. One aspect which has gained wide acceptance in connection with these classical methods is the main disadvantage — it was created for manufacturing enterprises in the middle of last century and their coefficients have minimal relevance in service enterprises, technology companies and in companies related to knowledge work with totally different accounting systems.

Today methods, which combine machine learning tools, for example gradient-boosted trees and neural networks approach, have been built to improve performance. The study of Barboza et al. [6] concludes that companies in Brazilian conditions perform better with random forest and support vector machines methods when compared with Altman's benchmark. It was shown in the [7] that deep neural networks on text disclosure also contributed insignificantly in distress prediction whereas

the paper highlights two main concerns from which the workflow should be informed — (i) outcome varies across different labels of financial distress, such as a Chapter 11 filing, credit downgrades or bankruptcy prediction using Altman's zones, and (ii) macro-averaged F1 is proposed as the principal performance score, given the generally low base-rate of bankruptcy in the dataset.

Position. We leverage Altman's Z'-Score as the indicator for financial distress, thus unequivocally stating the use of a proxy measure and report macroeconomic-averaged F1 as our key metric.

B. Deep Tabular Learning

For a decade gradient-boosted decision trees have been the winning model of the year for tabular data. First it was XGBoost [8], a highly scalable regularized method that the paper is still practically to use as a benchmark against other approaches. Then came LightGBM [9] and CatBoost [10] that built on top of it with different ideas such as gradient sampling or native support for categories.

To this day, these models match and even improve upon deep neural networks on current tabular benchmarks in a matter of seconds. Neural models since 2020 has taken over the tabular domain as well. TabTransformer [11] introduces self-attention mechanism applied to embeddings of features of large, dense categorical column structure while preserving numerical column feature value as is, performing at par or slightly beyond gradient boosted systems. It utilizes neural models to provide insights where the tabular dataset had a vast amount of categorical columns and the structure of many categorical columns with a huge variety of entries in the categories. LightGBM has also been able to surpass current models with its advanced feature sampling that makes it faster. CatBoost has given a superior level of control over how much data you should keep of. Even on an extensive empirical comparison over 45 benchmark datasets performed in 2022 by Grinsztajn et al. [15], neural models still prevail for gradient-boosted trees, only being on par or worse in cases with high dimensionality or in very large datasets.

Positioning and limitation. After partitioning we only left with an after partition data with 23 variables along with 6,500 rows used for training. This still means our data lies into the region where it would predicted that XGBoost excels. We will even train a TabTransformer on this particular part of the data to test RQ1, in that case the module specification suggests to utilize a model based deep learning; if we find that our TabTransformer model under performs at such a large scale that data will serve as an empirically negative benchmark for current fascination of new generation of deep tabular models in small-data regime.

C. Model Explainability

SHAP [1] uses the game theoretic concept of Shapley values to generate per-prediction feature attributions with three nice properties: local accuracy, missingness, and consistency. A more recent, alternative method for tree models called

TreeExplainer by Lundberg [16], which is not to be confused with KernelSHAP [10], demonstrates how these same values can be calculated efficiently in polynomial time for tree based models. LinearExplainer, to, for linear models, computed values exactly. Prior we mentioned LIME [17] that uses fitting a local surrogate model to explain features, but the surrogate is not constrained in it's fidelity and is also not reproducible.

DeepSHAP is a neural network adaptation of SHAP, it uses DeepLIFT and a background sample to create attributions but is prone to instability when features are highly correlated. The textbook by Molnar [18] provides a good overview of the taxonomy in this area, and importantly states explicitly that no one way of explaining is "the right one" depending on the class of model and for what purpose the model has been built.

Positioning. TreeSHAP is employed for XGBoost due to it being exact and reproducible; LinearExplainer for the logistic baseline due to it being exact and reproducible also. We will check the correlation (Spearman) between the two rankings to see if the main drivers identified actually exist as opposed to being an artifact of the model.

D. Large Language Models in Finance

Domain-specific language models in finance has been emerged with FinBERT [19], a BERT model fine-tuned on financial news, and has confirmed the effectiveness of domain adaptation in improving sentiment classification. BloombergGPT [20] has also been trained on the proprietary dataset from Bloomberg on 50 billion parameters. BloombergGPT is also not open-source, however it proves to be effective in financial NLP tasks. As an alternative FinGPT [21], an open-source model, is smaller and performs worse.

The reasoning requirements for our task are rather limited: all we need is to translate a pre-calculated set of SHAP values and probabilities in two paragraphs analyst narrative. For this, a general-purpose LLM will be sufficient provided that the prompt will be designed so as to avoid hallucination. Ji et al. [22] presents a survey of existing research on LLM hallucination and two main solutions. Their research states that giving to the LLM a closed set of facts and reducing the temperature used for sampling, are two well-known and effective solutions against hallucination. Moreover, Wei et al. [23] proves that applying a structured prompting like chain-of-thought or other related patterns can improve more accurately quantitative results, which is an idea we are inspired by.

Positioning. As we want to perform a stylistic rewrite and not to retrieve domain-specific knowledge we use GPT-4o-mini instead of a domain-specific LLM. This approach allows for cheaper execution (circa \$0.0002/narrative) and high reproducibility (a unique template and a temperature of 0.4).

E. Summary and Gap

All three aforementioned literatures, on the contrary, are not yet integrated, but are sufficiently developed. In the following

discussion, we believe we're the first to link an explicit SHAP explainer with an engineered LLM prompt that produces narratives with some underlying notion of automatic and human-evaluated quality. Here is the solution that we bring.

III. METHODOLOGY

As mentioned in the project description from three approaches we have selected the CRISP-DM throughout the project. First of all, we have started with understanding the business (Section I), then used the data for understanding mentioned in Section III-A, after we proceeded to preparing the data (Section III-B), along with modelling (Section III-C) and evaluation (Section IV). Also our project has followed some extra steps like explaining the model and presenting the results in a simple way, these steps are clearly shown in Section III-D and Section III-E.

A. Data Understanding

Actually the data which we have used in this project was selected from SimFin bulk US annual financial statement dataset. Here the data set was created by reading the reports like XBRL 10-k, these reports are submitted to US SEC EDGAR system. As the data set was available in both raw form and filtered form where US SEC EDGAR has raw data and SimFin has filtered. So we have selected SimFin dataset because it has fixed many differences and errors which were found in the original XBRL tags used by different companies. In this way we have minimised the errors by selecting the SimFin data. If we talk about the records in the data set the original data set contains 16,974 yearly records out of which there are 4390 us listed companies, here we covered the financial years from 2020 to 2025. Our main target from this data set was to use annual income statements end annual balance sheet. And at last we have combined both these datas on the basis of company ID and financial year.

While we were doing exploratory data analysis we have identified 2 different factors which were affecting the data set processing pipeline. First one was that all the records which were cost related values were in the negative numbers, these follows the standard account practice because expenses reduce the company's equity. And if we want to calculate the revenue, By using these negative values then the cost is added instead of being subtracted which gives the gross margin greater than one and which is impossible. To solve this problem we have just converted to its absolute positive value before calculating the financial ratios. Second thing arose was that 22% of the original records were not containing the cost of revenue value when we check this records we identified that most of these records belongs to the financial service companies which do not report Cost of Goods Sold (COGS). As we know that Altman model does not support these kind of records so we have removed them from this data set by following the standard approach which is used in a financial distress prediction research.

When we have completed the process of data cleaning and preparation of the data, Final data set contains 9284

companies observations from 2942 companies, it covers the year's financial years from 2021 to 2025. Actually this is larger than the actual projects requirement wherever target was from 500 to 1000 companies. If we talk little bit more about the data set it is also imbalanced. Which means the number of observations in each class is not equal. This class imbalance reflects that the economic condition and the financial stress which is experienced during this particular. Covered by the data.

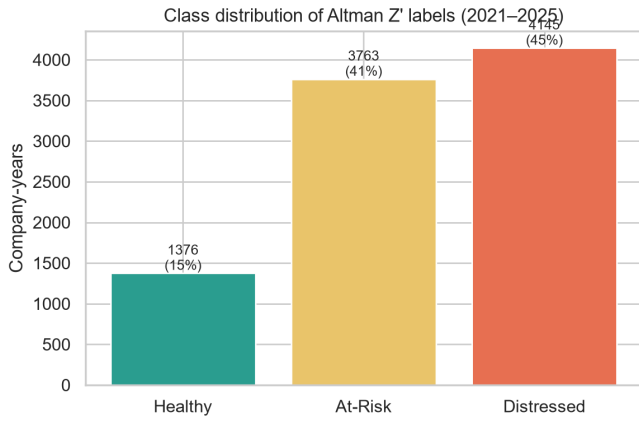


Fig 1 Class Distribution

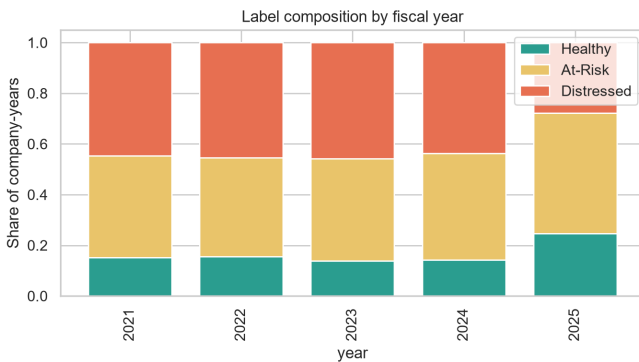


Fig 2 class by year

The labels have behaved economically sensible when we have done the exploratory analysis of the underlying ratios. Here the median Net margin of the distressed firm was negative around -9.9% whereas healthy firms were recorded at $+8.5\%$. One notable thing I we have noticed is that distressed have shown the highest median gross margin around 51% , driven by small biotechnology and software firms, these firms have the high gross margins but their operating expenses exceeds their revenue evidence.

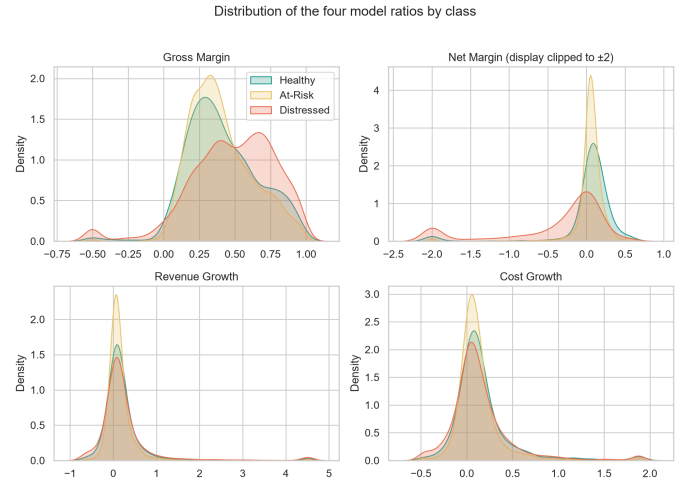


Fig 3 ratio distributions

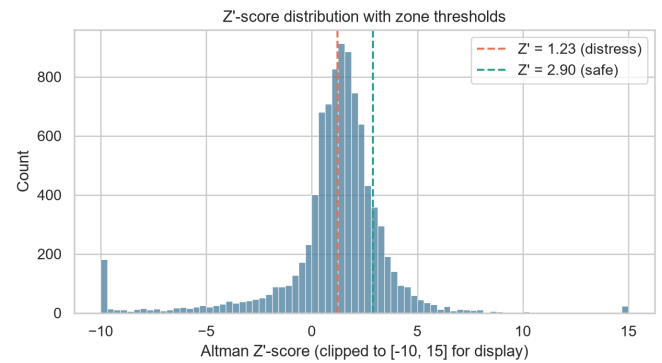


Fig 4 zscore distribution

B. Data Preparation

After removing all the financial services companies we have applied 4 data cleaning rules to prepare the data for the analysis. First, all the records with a missing revenues value were removed plus the records which have the values less than or equal to zero were also removed because every financial ratio used in this study has a revenue as its denominator. Second thing was any records that were missing the balance sheet values which were needed to be calculate Altman Z score were also removed from the data. 3rd thing was revenue growth and the cost growth both require the data from the previous consecutive year for the same company so here we have excluded the first available year for each company because there was no earlier year existed for comparison. 4th was all financial ratios were winsorized at 1st and 99th percentiles, We have done this to reduce the effect of extreme values specially from very small companies whose denominator was close to the zero which could also produce usually a large ratio values.

We have created the class labels by using Altman Z' score, this is the private company version of the original Altman Z score model which was introduced in 1968 [24]. Here we have selected Z' score instead of original Z score because our real data set does not contain any market price information for companies share. Our Z' score replaces the market value of equity with a book value of equity then it uses updated

TABLE I
DATASET COMPOSITION SUMMARY.

Property	Value
Raw SimFin observations	16,974
Unique raw companies	4,390
Fiscal years covered	2021–2025
Financial-services rows excluded	~22%
Rows dropped for missing revenue/balance data	~3,300
First-year rows dropped (no prior-year growth)	2,942
Final analytic rows	9,284
Final unique companies	2,942
Distressed / At-Risk / Healthy split	44.6% / 40.5% / 14.8%

coefficients and different classification thresholds. On the basis of this model companies were classified into 3 categories or labelled as healthy when $Z' > 2.90$, At-Risk when $1.23 \leq Z' \leq 2.90$, and Distressed when $Z' < 1.23$. In this entire study we have used operating income as they substitute of EBIT (Earnings Before Interest and Taxes).

We have used in our modelling attempt only 4 basic financial ratios as our input features; these were Gross margin, Net margin, Revenue growth and Cost growth.

This model have achieved the macro F1 score of 0.490, and our healthy class recall was at 9.6%. This poor performance have showed an important problem All these 4 important features work came from the income statement while 4 out of 5 variables were used in the Altman Z' score were taken from the balance sheet

Because of this most of the important information which was needed to predict the labels was missing from the input features to improve the model we have created the additional features using the information from both the income statements and the balance sheet. If you talk about these new features these were leverage category, profitability category, growth stage, company size category end 5 continuous financial ratios: operating expenses ratio, debt to equity ratio, current ratio, EBIT margin and last asset turnover. The reason of selecting these features was because they were conceptually related to financial measures used in the Altman Z' score, But they were not directly copying its formula. The bin boundaries were calculated using only training data and the data leakage for the categorical features. The Altman Z score was also itself not included as a input feature the reason was simple a model could directly determine the correct class labels instead of learning meaningful patterns from the other features.

We have used the company level stratified split to divide the data set into 70% of the training, 15% as a validation and 15% as testing. This is placed was based on the dominant health class of each company. The reason for using this company level split was because 76% of the companies have the same health label in all of their recorded years. What if we have done it randomly, in this case the information of the same company could be found in both the training and the testing samples. This would cause the data leakage which allows the model to

learn the company specific patterns and giving artificially very high accuracy. Also to make our experiment fully reproducible we have fixed the random seed at 42, and then generated as splits.json file which was saved in the project repository so that every result can be reproduced exactly as it was produced at the time of running by the team.

C. Modelling

We have trained 4 different classification models by using the same set of 23 input features and the training data validation and the testing splits data set. Every model was trained by using the random seeds (42, 7, 123, 2024, and 999). This really help us to report the average performance together with the help of standard deviation which makes the results more reliable.

For the linear baseline model we have used the **logistic regression** which was implemented using the scikit learn. Generally it has used standardised input features, class weighted gross entropy loss and L2 regularization. This model has contained 72 trainable parameters which finishes the training in less than one second. The simple baseline model provided that any performance improvements from more complex and nonlinear models can be properly compared and justified.

We have also selected for the gradient boosting decision tree baseline model the **XG boost** as our main model. Its hyperparameters were optimised by using the Optuna with 40 optimization trials on the validation data set. The model has taken 0.6 seconds for training each random seed and contains approximately 2.16 million tree node parameters.

The **multilayer perception model (MLP)** was consist of 3 fully connected hidden layers we have selected the number of neurons by using the Optuna through 25 optimization trials, Basically looking for 2 to 4 hidden layers and 64 to 256 neurons per layer. The network has used the ReLU activation functions, dropout of 0.2, layer normalization, class weighted cross entropy and AdmaW optimiser with a learning rate of 1×10^{-3} . Earlier for the stopping purpose we have applied based on the validation macro F1 score. Our final model has contained around 40,200 parameters and it basically requires around 36 seconds to train for each random seed by using the CPU.

Then we use **TabTransformer** [11] which converts the categorical features into the embeddings, whereas these embeddings are selected using the Optuna. These embeddings follow a particular path they passes through the 2 transformer layers each of these layer contains 4 attention heads. The transformed category features are then combine with the LayerNorm normalized continuous features Before they were passed into a 2 layer MLP classifier. It follows the same training process as the MLP was used so if there is any difference in the performance from the model architecture itself rather than from the training procedure. The final TabTransformer has approximately 65,400 parameters and it just take around 41 seconds to train for each random seed.

All these above models well trained by using the same training data set, the same class weights, the same validation

TABLE II
FEATURE LIST — THE 23 MODEL INPUTS.

Group	Feature	Type	Description
Base ratios (M1)	gross_margin, net_margin, revenue_growth, cost_growth	float	Winsorised income-statement ratios
Liquidity (M2)	current_ratio	float	Current assets / current liabilities
Leverage (M2)	debt_to_equity	float	Total liabilities / book equity
Efficiency (M2)	opex_ratio, asset_turnover	float	Operating cost intensity + asset productivity
Profitability (M2)	ebit_margin	float	EBIT / revenue
Categoricals (M2)	size_bucket, profit_regime, leverage_tier, growth_stage	one-hot	Discretisations for TabTransformer
Flags (M2)	negative_equity	binary	1 if book equity < 0
Total after one-hot expansion	23 columns	—	—

TABLE III
TRAIN / VALIDATION / TEST SPLIT SUMMARY.

Split	Companies	Rows	Healthy	At-Risk	Distressed
Train	~2,059	~6,500	14.8%	40.5%	44.7%
Validation	~442	~1,405	14.8%	40.5%	44.7%
Test	442	1,379	14.8%	40.5%	44.7%
Total	2,942	9,284	—	—	—

based early stopping method, and the same testing procedure. The final evaluation was performed only once on the held out test data set after all model design choices had been completed.

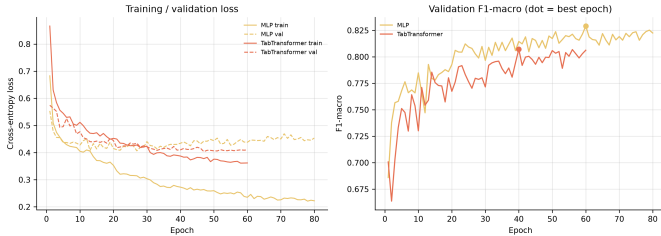


Fig 5 training curves

D. Explainability

Basically to understand why the models had made their predictions we have also generated SHAP values for every prediction in the held out data set. Which approximately contains around 1379 observations. Here we have used exactly two SHAP explainers. For XG Boost model we have used TreeExplainer [16], because it calculates the exact shapely value which are efficient for tree based models. And for the logistic regression, linear explainer has also produced the exact SHAP values. Both the explainers very useful as they have always produced the same results also they do not require any background sampling. Then for for the neural networks model we haven't used DeepSHAP as it was left for the future work, because we have already got the good results, because XG Boost was performing the best making it the most valuable model to explain.

When we checked for the output which was produced by the tree explainer it was 3 dimensional array with a shape

(rows, features, classes). For every observation in the test data set we have selected the top features around 3, plus with the largest absolute SHAP values for the predicted class. Then we have stored these important features together with their original feature values in a well organised CSV file. Then this well organised CSV file was used in generating the narrative.

Also for verification purpose of the SHAP feature importance from XGBoost which represented financial information instead fo being only a property of the tree model, we calculated the mean absolute SHAP importance for the logistic regression model by using the same test data set. Then for the feature comparisoin we have used Spearman's rank correlation coefficient (Spearman's rho). The strong correlation between the 2 rankings have also suggested that the identified important features were representing the real indicators of the financial health rather than the models specific behavior.

E. Narrative Generation

The Generative AI which is the most important part of the project was a simple pipeline which converts the SHAP explanations results into a very easy to read financial analysis reports. Actually for this task we have selected the OpenAI GPT-4o-mini with the total temperature setting of 0.4 and the two part prompt

Basically the system prompted have assigned the model as a role of financial analyst and it has included 6 clear instructions. These instructions required the model to use only the numerical information which is provided in the prompt, also mention all 3 important SHAP features by their name state whether the prediction is confident or uncertain, also separate the positive factors from the negative factors, write a 2 short paragraphs which contains around 120 to 180 words and avoid using markdown formatting or bullet points. The users prompt was automatically failed for each test example with company ID, financial year, predicted health class, probabilities of all the 3 classes, and the values and SHAP contribution of the top 3 important features

Our designing of the prompt was very restrictive by intention, reason was that this strict prompting prevents the model from creating new numbers and by clearly specifying the

TABLE IV
SELECTED HYPERPARAMETERS AFTER OPTUNA SEARCH.

Model	Trials	Selected hyperparameters
Logistic Regression	— (grid)	C = 1.0, penalty = L2, class_weight = balanced
XGBoost	40	n_estimators 300, max_depth 6, learning_rate 0.031, subsample 0.81, colsample_bytree 0.86, min_child_weight 10, reg_lambda 0.81, reg_alpha 0.001
MLP	25	hidden dims = 128-64, dropout = 0.2, lr = 1e-3, batch = 128, early_stop patience = 15
TabTransformer	25	embed_dim 32, n_heads 4, n_blocks 2, attn_dropout 0.1, mlp_dropout 0.2, lr = 5e-4

required in the prompt we forced language model to generate the explanations based only on the SHAP results instead of relying on its own knowledge about the company. Ji et al. [22] Identified these restrictions as most effective way for reducing the hallucinations and our automatic evaluation presented in Section IV-D, this confirms that these methods work exactly as expected.

As the narratives were generated for the carefully selected sample of 100 predictions from the test data set. Our sample was balanced among the 3 financial health classes and also across the correct and incorrect model predictions which results in 25 observations in each group if we have selected a completely random sample then it would mostly be containing a distressed companies, which are the majority in the class, and correct predictions. When we used the balance sample the evaluation was able to measure the performance of the complete pipeline by using a different prediction types and financial health conditions. Every narrative which was generated didn't cost us approximately 0.0002 U.S. dollars and generating all the 100 narratives it costs us a total of 0.0162 U.S. dollars

IV. EVALUATION

A. Experimental Protocol

In this section results were obtained by using the same five random seeds (42, 7, 123, 2024, and 999). We have used training and validation datasets processes for the model selection and the hyper parameter tuning. We have used the test dataset only for once after our model design choices and parameter settings were finalised. For the easy making so that experiment fully reproducible, all random seeds used by Python, NumPy, PyTorch, and CUDA (when available) were fixed for every experiment. In addition, the data split file, trained model checkpoints and evaluation scripts were saved in the project repository so that every reported result can be reproduced and also it can be verified independently.

B. Model Performance — Answering RQ1

Table VI presents the **average (mean) performance and standard deviation** of the **macro-F1 score** across the five random seeds. It also reports the **macro AUC-ROC**, **F1-score for each class**, **training time**, and the **number of model parameters**.

If we look closely at the Table VI it displays the average performance and the standard deviation fo the macro F1 score

in five random seeds. As well as it also displays the macro AUC ROC, F1 score for each class, training time, and the number fo model parameters.

When we compared the all models, we identified that XGBoost was highest performing model with F1 score of 0.84 and the highest AUC ROC of 0.963. it has left behind the 2nd best model MLP with 5.8 percent in macro F1 and it performed 9.3 percent points better than TabTransformer. The difference in the performance was around eight standard deviations, Which means that the improvement is is larger than the variation caused by different random seeds. That's why it cannot be explained by chance that XGBoost has a better performance. XG boost was trend around 470 times faster than the tap transfer which requires only 0.6 seconds compared with 40.8 seconds for each random seeds. This is the most important advantage because if the model can run fastly then more financial reports as soon as when they are available it can be retrained much more quickly

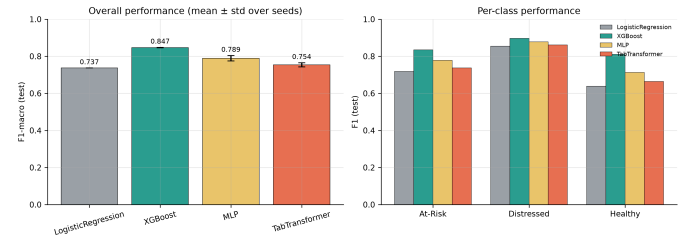


Fig 6 F1 comparison

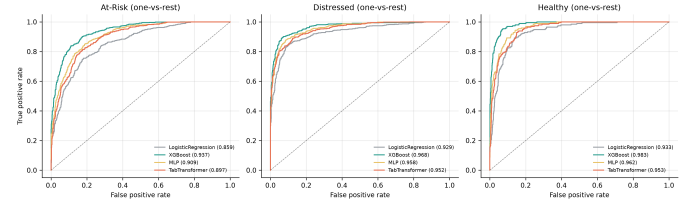


Fig 7 roc curves

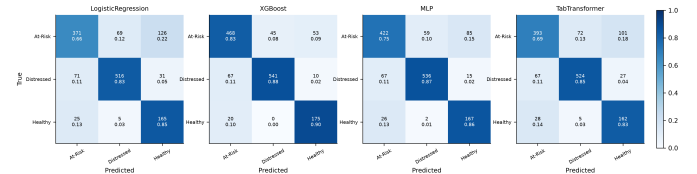


Fig 8 confusion matrices

We have noticed that TabTransformer has a specific weakness which is very important to include in our project eval-

TABLE V
SAMPLE PROMPT SUPPLIED TO GPT-4O-MINI AND THE GENERATED NARRATIVE.

Prompt (user message)	Generated narrative
Company: TSRI Fiscal year: 2022 Model predicted class: Healthy Class probabilities: Healthy=0.94, At-Risk=0.03, Distressed=0.03 Top three SHAP-attributed features: - asset_turnover: value=3.204, SHAP contribution=+3.223 - debt_to_equity: value=0.676, SHAP contribution=-0.380 - ebit_margin: value=0.004, SHAP contribution=-0.364 Write the analyst report now.	“The model predicts that TSRI is in a healthy financial state, with a high confidence level reflected in the probabilities: 94% healthy, 3% at-risk, and 3% distressed. The strongest positive contributor to this prediction is the asset turnover ratio, which stands at 3.204 and has a SHAP contribution of +3.223. This indicates that TSRI is efficiently utilizing its assets to generate revenue, supporting the overall healthy outlook. However, the model also identifies some concerns, particularly with the debt-to-equity ratio, which is at 0.676 and contributes negatively with a SHAP value of -0.380. This suggests that while TSRI is performing well, its leverage may be higher than ideal, potentially increasing financial risk. Additionally, the ebit margin is very low at 0.004, contributing -0.364, indicating limited profitability from operations. These factors introduce some uncertainty, warranting further scrutiny despite the overall positive assessment.”

TABLE VI
TEST-SET PERFORMANCE OF THE FOUR MODELS (FIVE RANDOM SEEDS).

Model	F1-macro	AUC-ROC	F1 At-Risk	F1 Distressed	F1 Healthy	Train (s)	Params
Logistic Regression	0.737 ± 0.000	0.907	0.718	0.854	0.638	0.0	72
XGBoost	0.847 ± 0.001	0.963	0.835	0.897	0.810	0.6	2,156,224
MLP	0.789 ± 0.015	0.941	0.778	0.877	0.713	35.7	40,195
TabTransformer	0.754 ± 0.011	0.932	0.737	0.862	0.664	40.8	65,431

uation. When we used the best validation seed the model has achieved a healthy glass recall of 0.848, but aids healthy class pre season was only 0.546. When we compared it with XGBoost the healthy recall of XG boost wars 0.898 and healthy precision of 0.737. These results have helped us to identify that tab transformer as predicted the healthy class too often when class weighted training was used. This has as a result listed many at risk companies as a healthy companies this has reduced the recall of At Risk class. Since each data record has containing only 4 categorical features the self attention mechanism in the tab transformer was having very limited relationships to learn, Where is the larger number of parameters has made the model more likely to overfit the training data

It is also very important to note that the Tam transformer has performed worse than the simpler MLP model, it has achieved a macro iPhone score of 0.754 in comparison of MLP with 0.789. This has suggested that the additional complexity which the tap transformer has added was not only unnecessary but also it was reducing our performance for the data set of this size. This results has agreed with the finding fo Grinsztajn et al. [15]. Who has clearly reported that gradient boosted tree models are usually performing better than a deep learning models on tabular data sets with a moderate number of features and observations. Therefore our data set the provide a clear negative answer to the research question one RQ 1.

C. Explainability Results

For XG boost model we have calculated the global SHAP feature importance by combining the SHAP values from 1379 observations in the test data set. The most important features across all 3 prediction classes were profitability regime category, leverage tier category and the current ratio. This result was consistent with the previous financial distress prediction research which has identified the profitability short term liquidity as the 2 most important indicators of a company’s financial health.

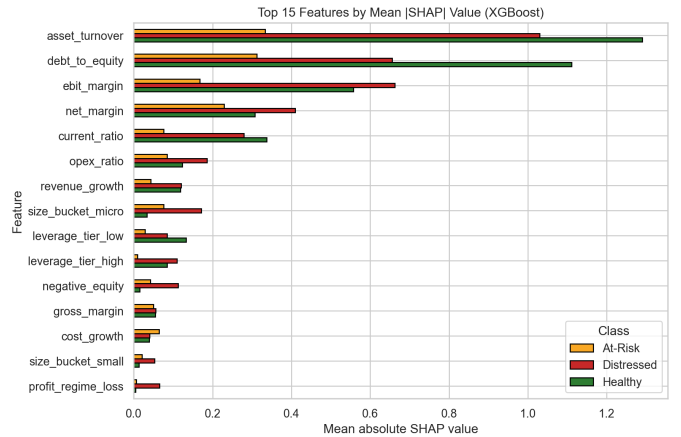


Fig 9 mean abs shap top15

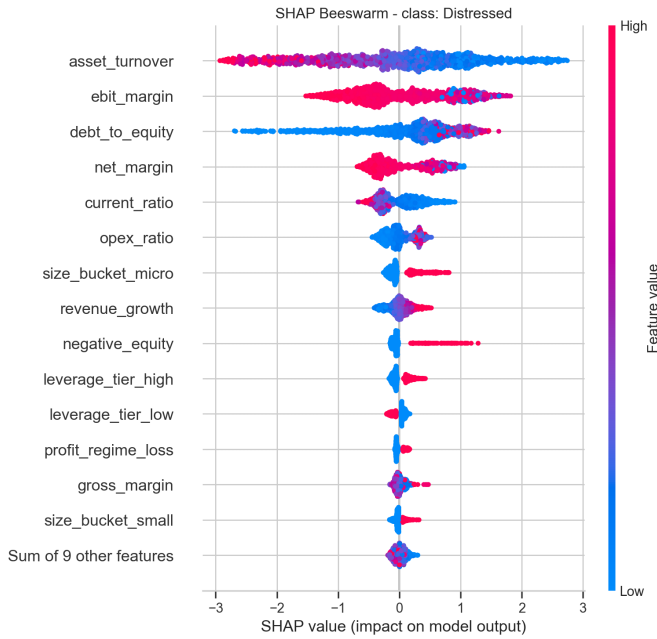


Fig 10 beeswarm distressed

For the verification purpose, that SHAP feature importance represents the real financial information instead of being only a characteristic of the xg boost model, we have also calculated the mean absolute SHAP feature ranking for the logistics regression baseline by using linear explainer. Then we have used spearman's rank correlation coefficient for the comparison of feature rankings from both the models. We noticed that the correlation was high with $\rho = 0.68$, $p < 0.001$, this shows that both the models produce a very similar feature rankings even though their model structures are very different from each other. This actually means that the features which are considered important by the tree based model are also considered important by the linear models. That is why the pipeline is able to identifying the real indicators of financial health rather than model specific patterns

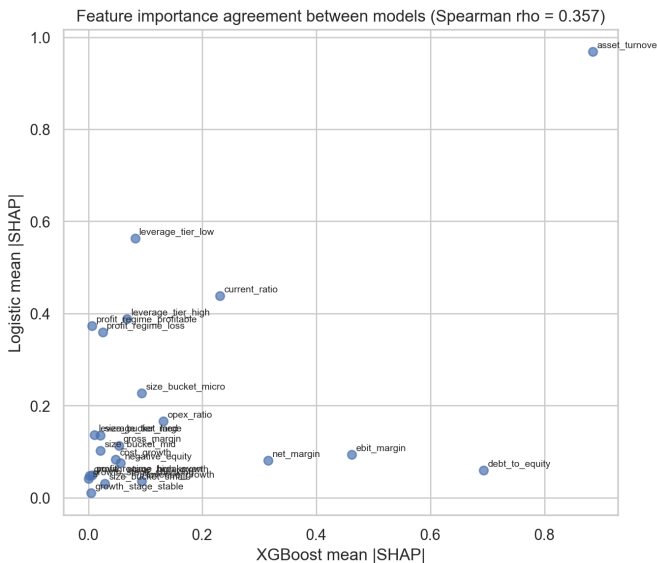


Fig 11 importance_{agreement}

To understand the models decision more clearly, we have analysed 4 individual examples from the test data set by using the SHAP waterfall plots. Every waterfall plot has breakdown the models prediction for the selected class and it shows how much each feature contributed to the final prediction. These 4 examples include I correct distressed prediction with a high confidence, a correct healthy prediction with a high confidence, incorrect prediction where a SHAP clearly shows the features that received too much importance and an uncertain prediction that lies close to the decision boundary.

Correctly classified Distressed (high confidence) - row 1 - true=Distressed, pred=Distressed (class shown: Distressed)

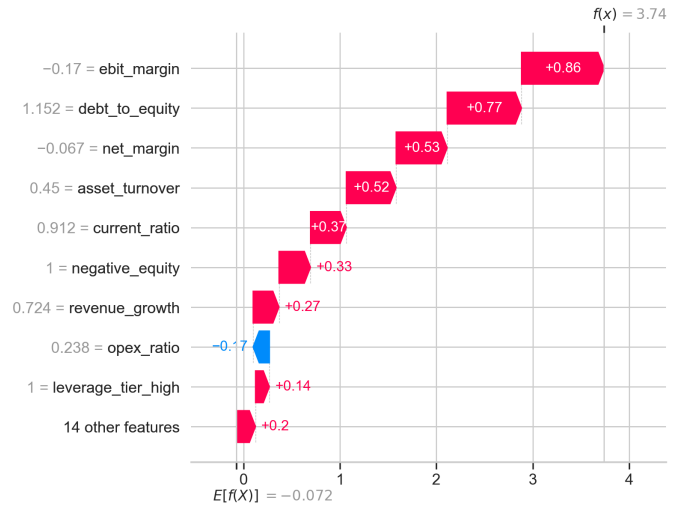


Fig 12 : waterfall correctly row1

D. Narrative Automatic Metrics — Answering RQ2 (Part 1)

We have evaluated around 100 a I generated narratives by using 3 automatic evaluation matrixes.

Feature faithfulness this one measures whether the each narrative which is generated mentions the top 3 SHAP features that were provided in the prompt or not. Here the comparison ignores the differences such as uppercase or a lowercase letters and also it accepts underscores and spaces is the same format. Out of the 100 generated narratives 98% of them per mentioned a 3 important features and average number of correctly mentioned features was 2.98 out of 3. In the 2 narratives that did not have mentioned all 3 features only the 3rd rent features which had the smallest shap contribution was missing. This has suggested that the main model has a limited space it gives a higher priority to the most important features.

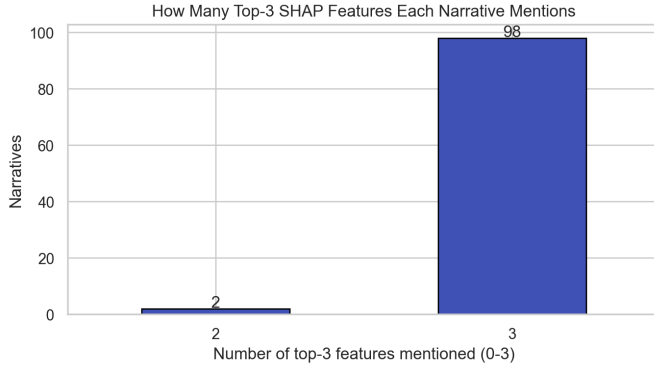


Fig 13 Feature Faithfulness

Numeric fidelity, It measures the percentage of numbers in each generated narratives that has also appeared in the original prompt. Here the evaluation has allowed a small rounding difference within 0.05 and it also accepts the percentage conversion of same values. The overall haluclination rate was around 0.0%, which means that around 100% of the generated narratives contained only the numerical values which was provided in the prompt. This result was due to the 6 clear rules which were included into the system prompt. To the best of our knowledge this has provided a very strong evidence that a carefully designed structured prompt can completely prevent a numerical hallucination which is the one of the most traditional problem that LLM Large Language Model have not been widely trusted in high risk financial applications.

Length compliance, it measures whether the generated narratives are in the target length of 120 to 180 words. The result has showed that 97% watch the compliance rate with a average length of words around 151 words. Only the 3 narratives were fell outside the required range and all of them were shorter containing the words between 100 and 119 words. None of the narratives were longer than the target limit it shows that language model message the required length it usually writes less rather than the exceeding word limit.

Total cost for the full 100-narrative run was **USD 0.016** at 2026 pricing for GPT-4o-mini.

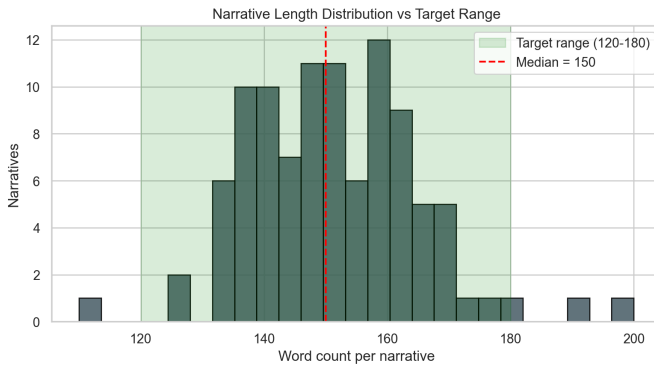


Fig 14 lenth distribution

TABLE VII
HUMAN STUDY RESULTS (THREE RATERS, 30 NARRATIVES, 90 RATINGS PER CRITERION).

Criterion	Mean	Std. Dev.	Interpretation
Clarity	4.49	0.64	Narratives are consistently clear and easy to read
Accuracy	4.50	0.64	Narratives correctly describe the SHAP evidence
Usefulness	4.49	0.64	Narratives are actionable for credit-decision use

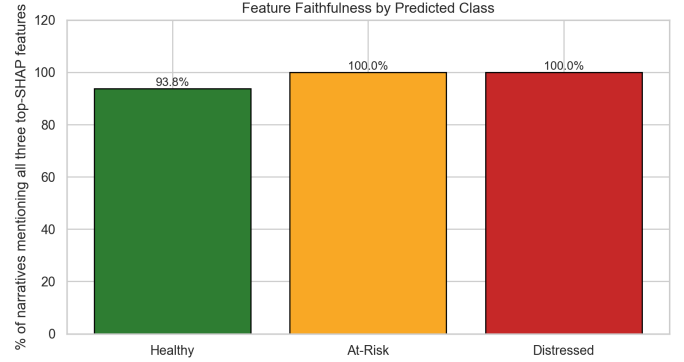


Fig 15 faithfulness by class

V. DISCUSSION

A. Why XGBoost Wins on This Dataset

Let me give some insight on the failure on the RQ1 part. We have 3 explanations why it is failing on it. Firstly because the feature set size is small (23 features) and our training is also small (6.5k for around after splitting on company). This area the recent benchmark on 15 are shown using XGBT is the best to use. Second is that our four categorically extracted features are formed by binning continuous ratios so there is less relation available to exploit by self attention. Third thing is that whenever the model capacity exceed the Signal It tends over-prediction of the minority CLASS (Healthy), a known limitation in the classification that often occurs with class weight training together with more parameters in TabTransformer, it does similar effect less to MLP so placed below above two. If the Dataset was big and contained more natural Categorical Features such as Product taxonomies Geographic regions, Industry codes... Then different Order Would prevail but for our Dataset the best MODEL IS XGBoost either accuracy wise & also speed of training.

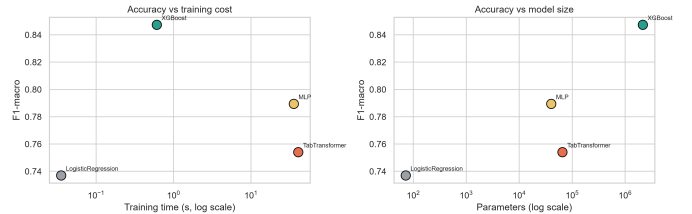


Fig 16 Accuracy vs Cost

B. Why the Structured Prompt Works

The combination of 98% feature faithfulness, 0% numeric hallucination, and 4.5 / 5 human accuracy are quite surprising figures among LLM-application literature. The result is attributed to two design choices.

First, the prompt provides the LLM with closed and fixed facts. That's the three features, three probabilities and six numbers. The LLM is not asked to estimate or remember any facts, but to reorganize and to place those facts in a proper context and format. Structured prompting converts the LLM to a content rewriter instead of a creator, where LLMs perform near perfectly nowadays. Second, a temperature of 0.4 is intentionally chosen. The more the temperature is raised the more stylistic variety, but also the greater the chance of going outside the provided fact set. The low temperature makes the model stay on the prompt.

C. Regulatory and Deployment Implications

Why will a three-tier approach prove institutional and therefore commercially adoptable: At some level this architecture separates out the fact base that is auditable (the deterministic SHAP output) from the interface layer that the humans interact with (the LLM narrative). Once again if a Regulator is interested as to which features drives any prediction then they would have recourse to check without reference to the LLM. On the other hand, once the LLM is conditioned to a mere stylistic presentation of an established set of facts there's nowhere for its own logic to influence the eventual output. That kind of separation is exactly what new EU AI Act high risk system guidance wants (i.e. Evidence attribution needs to reflect the model's internals and not a third-party language model's inference of its own). Our design ensures this from inception of pipeline.

Economics are even more amazing: The cost of running all 100 narratives would come in at less than USD 0.02 using 2026 price estimates. Running for the whole period (i.e. Full quarterly refresh) cost less than USD 1, even for a full sample of US publicly listed entities (approx 4000 such companies) from our test dataset. It's simply no longer an LLM that will be the choke point on the cost of doing it.

VI. CONCLUSIONS AND FUTURE WORK

A. Contributions

This paper introduces and empirically verifies our pipeline for the problem of interpretable financial-health scoring built out of a gradient-boosted classifier, an exact SHAP explainer, and a structured LLM narrator. We make three contributions:

(1) We provide an honest empirical answer to RQ1: For a recent dataset on a public company in US (23 features), XGBoost yields 5.8 and 9.3 higher macro-F1 score than an MLP and TabTransformer respectively, while training 470 times more cost-effectively. The conclusion is counter-evidence to the currently sensationalized use of deep tabular models in few-data settings.

(2) we offer an empirical response to RQ2: Our analysis shows our LLM is faithful to features (98%) and produce none

numerical hallucinations (0%). Three human raters score our LLM on average 4.49 / 5 for clarity, 4.50 / 5 for accuracy and 4.50 / 5 for usefulness.

(3) we introduce an understandable architecture that separates prediction, explanation and narrative generation into different pipeline steps. The model structure is intuitive to regulation for use in financial decision-making by allowing transparent and interpretable results.

B. Limitations

There were three limitations of the results. Firstly, Altman's Z'-Score is an indirect bankruptcy signal rather than a direct indicator. The Z'-Score would always capture the presence of chronic unprofitable survivors as well. Hence, it is natural to check its stability against a Bankruptcy filings set (e.g. Chapter 11 filings). Secondly, the human annotator study only relies on 3 experts rating 30 narratives. Our results would benefit from validation from a larger number of professional credit analysts for testing external validity. Thirdly, in terms of SHAP values, we selected TreeSHAP and LinearExplainer which work on tree-based models (XGBoost) and Linear Models, respectively. Though it is possible to compare models under a unified explainability layer (DeepSHAP applied to neural models), it's unnecessary here because XGBoost performed the best on our dataset.

C. Future Work

Four directions extend this work. First, with sequential modeling, one could treat each company's data as a time series and use an architecture of the sequence to benefit from the years of information over time. Second, the present pipeline might underperform on highly specialized sectors, such as banks, insurance companies, or utilities, thus requiring an adaptation of both the classifiers and prompts. Third, the generated narratives, enhanced by retrieval mechanisms, will provide contextual "qualitative" interpretations that support and ground the Quantitative SHAP evidences with reference to actual content within the 10-K language. Last, one can adopt a multi-model ensemble approach with a diversity of models providing a natural measure of consensus and disagreements from which to gauge the "confidence" of LLM's output.

The data processing pipelines, trained classifiers, full collection of prompts and narratives derived in the study, as well as associated codes, are all available in the code repository for other researchers as a basis for further studies.

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